Alvin Lionel Thumi Thooko

Software Engineering student

Nairobi, Kenya

Phone: +254745419147

Email: alvinlionel053@gmail.com

https://lionel-portfolio-one.vercel.app/

GitHub:
https://github.com/AlvinLionel

Summary

Software Engineering student with hands-on experience building full-stack applications and software systems. Skilled in both backend and frontend development, database design, API development, and deployment workflows. Experienced in developing REST APIs, integrating third-party services, working with relational and NoSQL databases, building security-focused applications and on manual and automated testing of finishes applications.

Tech stack

Languages: C/C++, JavaScript, TypeScript, SQL
Frontend: CSS3, Tailwind CSS, Bootstrap, React, Next.js
Backend: Node.js, Express.js
Databases: MySQL, PostgreSQL, MongoDB

APIs & Integration: REST APIs, API Integration, Safaricom Payment Daraja API

Security & Cryptography: Web Crypto, AES, RSA, ECDH/X25519
Tools: Git, GitHub, Vite, npm, prisma, figma
Testing and Quality Assurance: Postman, Jira, Playwright, Cypress, Jenkins, Manual Testing, API Testing, Test Case Design
Engineering: Requirements Engineering, Database management, Technical Documentation, Version Control

Projects

GitHub: https://github.com/AlvinLionel/My_Projects

LockBoxX- Browser-based security system | *React, TypeScript, Web Crypto API*

Built a client-side security platform for encrypting and decrypting multiple resource types and with native-password support for some resources.
Implements symmetric algorithms, public keys, key exchange algorithms and native-password protection techniques to secure resources.
https://lockboxx.vercel.app/

Full-Stack E-commerce Platform | Ongoing

| Node.js, Express, React, TypeScript, MySQL Developing a full-stack e-commerce platform for discovering home products, nearby stores and home improvement services.
Implemented user authentication, product catalog, shopping cart functionality and REST API endpoints with an integrated interactive map functionality.

Checkout System | JavaScript, Node.js, MySQL, Bootstrap, Safaricom payment API

Built a store checkout page with product and transaction data managed through MySQL Integrated Safaricom Daraja to support mobile payment and processing.

Static Server | Node.js

Built a custom static file server using Node.js for serving files over http.
Implemented event and error logging to monitor server activity and diagnose failures.

Club Website | Next.js, Tailwind css, Express, Postgres, Jenkins

Designed, built and actively maintain a website for the aws student builder club at the university
https://awssbgatcuk.pages.dev/sbg

Others

Built both static and lively websites both from scratch and using no code platforms.

Education

The Co-operative University of Kenya

Bachelor of Science in Software Engineering
Expected Graduation: 2028.